

Business Summary Report: Predictive Insights for Collections Strategy

Prepared by Tata iQ for the Head of Collections, Geldium Finance | Based on 500 customer records (16% delinquent)

1. Summary of Predictive Insights

Our EDA and predictive framework (logistic regression, Task 2) point to a few early risk signals. **Important:** the quick baseline model showed limited predictive power (AUC about 0.5) and none of the differences below is statistically significant yet, so treat them as indicators to test, not proven drivers.

Top 3 Risk Factors

- **Employment instability:** Unemployed customers are 19.4% delinquent vs 15.2% for others.
- **Business credit card holders:** 21.3% delinquent vs 14.5% for other cards.
- **Higher credit utilization:** customers using under 30% of their limit are 12.0% delinquent vs 16.8% above 30%.

Key Insights Summary

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Unemployed customers carry higher risk (19.4%)	Unemployed customers, especially with utilization of 50% or more (23.8%)	Employment_Status, Credit_Utilization, Income	Offer early, supportive outreach and flexible payment plans
Business card holders are the riskiest card type (21.3%)	Business card customers; 33% when also unemployed (small group of 21)	Credit_Card_Type, Employment_Status	Product-specific outreach and monitoring
Risk rises once utilization passes about 30%	Customers using 30% or more of their credit limit (83% of customers)	Credit_Utilization, Debt_to_Income_Ratio	Use utilization as an early-warning trigger

Watch-out: payment history (Month_1 to Month_6, Missed_Payments) is normally the strongest predictor of delinquency but shows no clear link in this data. Geldium should confirm how Delinquent_Account and Missed_Payments are defined before the model goes live.

2. Recommendation Framework

Based on the combined insight above, we recommend a controlled pilot. The reasoning is set out using the SMART approach.

- **Restated Insight:** Customers who are unemployed or hold Business cards and use 50% or more of their credit limit show a 20.7% delinquency rate vs 16.0% overall (82 of 500 customers).
- **Proposed Recommendation:** Run an 8-week early-outreach pilot (SMS/email reminder plus an offer of a flexible payment plan) for this segment, with a randomly chosen control group that receives standard treatment.
 - **Specific:** proactive contact and payment-plan offer for unemployed or Business-card customers with utilization of 50% or more.
 - **Measurable:** cut the 30-day delinquency rate of the treated group by 10% relative to the control group (about 20.7% to 18.6%); also track contact and plan take-up rates.
 - **Actionable:** uses existing SMS/email channels and Collections staff; the model score selects who is contacted, and agents handle plan requests.
 - **Relevant:** targets Geldium's goal of lowering delinquency through earlier, more precise intervention rather than broad manual handling.
 - **Time-bound:** launch within 4 weeks, run for 8 weeks, and review results at 90 days.
- **Justification and Business Rationale:** Reaching at-risk customers before they miss a payment is cheaper than recovery and keeps the customer relationship intact. Because the evidence is still early, a pilot with a control group tests the idea at low cost and tells us whether it works before scaling. **For stakeholders:** "We will contact a small group of higher-risk customers early with help and payment options, and compare them with a similar group we leave alone, to see if early support reduces missed payments."

3. Ethical and Responsible AI Considerations

This work affects customers' finances, so fairness and transparency matter as much as accuracy.

- **Two fairness risks and mitigations:** (1) Proxy bias: Employment_Status, Age (45-60 shows 20.0%) and Location (Los Angeles 19.6%) can stand in for protected or socio-economic characteristics, so groups could be unfairly targeted. Mitigation: keep Location and Age out of the model, and test outcomes by group using the disparate impact ratio (flag if below 0.8) and recall gaps. (2) Unreliable small segments: groups such as unemployed Business-card holders (21 customers) can produce false flags. Mitigation: use control groups, review thresholds regularly and require human review of borderline cases.
- **Explainability:** we describe a flag in plain words, for example "this customer uses most of their credit limit and has recent late payments", and show the top reasons for each score using coefficients (logistic regression) rather than a black box.
- **Responsible decision-making:** the recommendation offers support (reminders, payment plans) rather than penalties, and no customer is refused or charged automatically on a model score alone.
- **Other principles:** transparency (documented features and results), accountability (a named owner and quarterly bias and performance review), privacy (only necessary data, pseudonymised IDs) and validation of all GenAI-assisted outputs by analysts.